# Maintenance for Directed Energy Deposition 345

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**Class Outline**

Directed Energy Deposition Maintenance
Machine Components
Wire and Powder Feed System Components
Machine Configurations
Review: Basic DED Maintenance Considerations
Basic Cleaning Methods
Deep Cleaning Methods
Maintenance Tasks: Checks and Inspections
Maintenance Tasks: Recording
Maintenance Tasks: Data Capture
Software System Maintenance
Review: Cleaning, Inspection, and Data Maintenance
Regular Maintenance
As-Needed Maintenance
Other Maintenance Considerations
Final Review

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## Directed Energy Deposition Maintenance

Directed energy deposition (DED) is an advanced additive manufacturing (AM) process that can create near-net shape parts and repair existing parts. It is also the most commonly used AM method in hybrid manufacturing operations. DED commonly uses metal materials but can also use composites and ceramic-metal blends.

DED operators must achieve consistent results, or repeatability. In addition to proper setup and operation, you can improve repeatability through regular, scheduled planned maintenance and predictive maintenance (PdM). PdM uses data collection and data analysis to anticipate machine issues as part of preventative maintenance (PM).

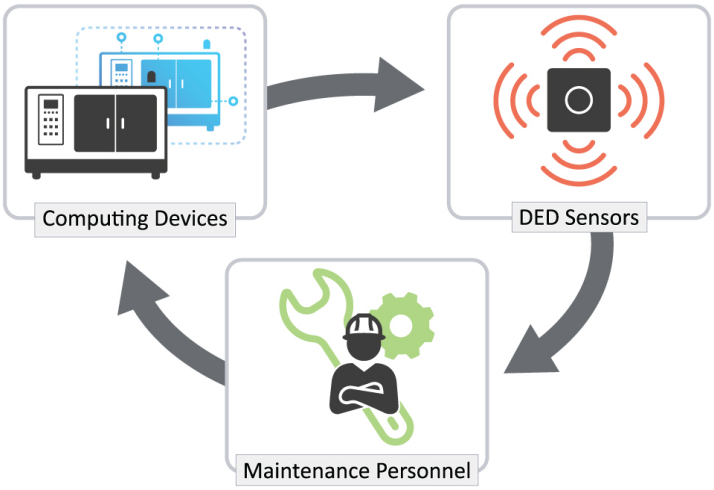


**Figure 1:**

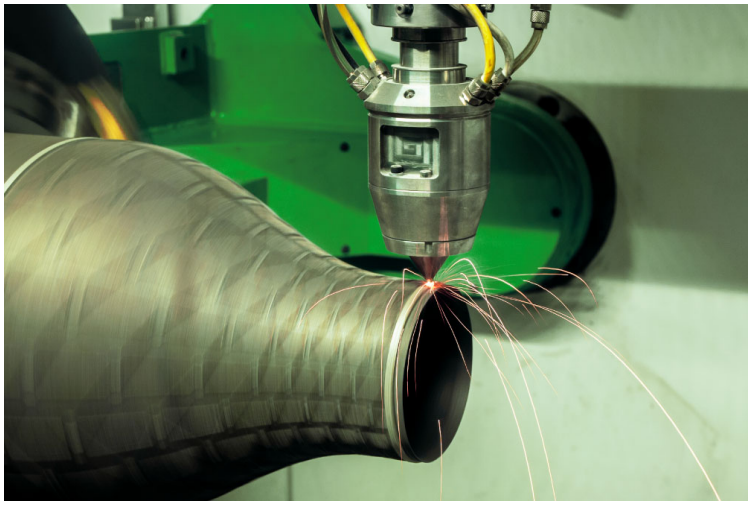

**Figure 2:** Photo courtesy of Add-Up

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## Machine Components

Maintenance tasks for DED typically involve the build plate and worktable, the energy source components, and the feed system components.

DED parts adhere to the build plate, so you must mechanically remove them. This process damages the build plate surface, so you must maintain the plate surface to ensure proper part adhesion and heat transfer.

You must also maintain components that produce the energy source to create the melt pool. For example, you must remove any spatter from the deposition head, nozzle, or welding torch. Spatter can affect the flow of gas or powdered materials, which can damage the next print and reduce component life.

Laser beam and wire arc energy sources require a shielding gas to prevent contamination. It is important to check pressure seals on shielding gas tanks and where gas is delivered to the machine. Pressure caps on seals may require tightening and must be repaired or replaced if damaged.

An electron beam requires an inert environment. Instead of shielding gas, it requires a vacuum-sealed chamber, which you must clean and maintain. You must also monitor and maintain environmental controls to prevent contamination, which can affect the integrity of the electron beam.

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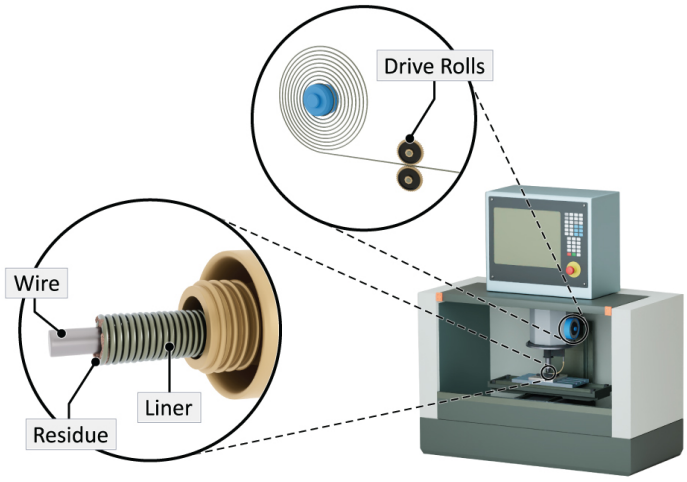
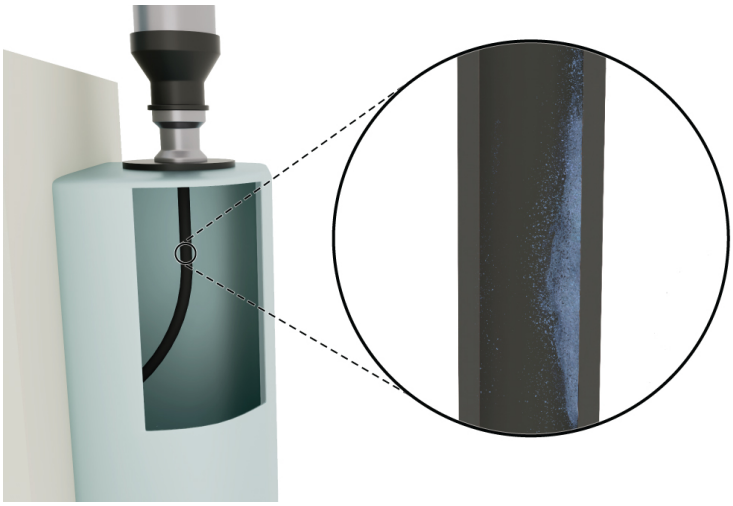
## Wire and Powder Feed System Components

While some advanced DED machines may be equipped with both, most machines use either powder feedstock or wire feedstock. Each system has specific maintenance concerns.




DED machines with wire feed systems force wire feedstock from a wire feeder through feed lines into the melt pool. Contaminants or excessive lubricant can clog or disrupt feed lines. You must ensure that the wire feeder and feed lines are clear and operating correctly.



DED machines with powder feed systems force powder feedstock from material hoppers through feed lines into the melt pool using a carrier gas. Powder feed systems also use a build chamber to contain any loose powder. You must maintain material hoppers to prevent contamination.

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## Machine Configurations

Different DED machine configurations may require different maintenance tasks.

Some three-axis DED machines use a stationary nozzle and a build platform that moves in three linear axes. Others may have a stationary worktable and a nozzle on a movable gantry. Creating or repairing parts with complex features on three-axis machines requires support structures, which requires technicians to clean components used for support materials.

Some four-axis machines combine a three-axis nozzle with a worktable that moves the workpiece in a fourth rotational axis. You may need to test the function of both linear and rotational axis movements to ensure that all gears, belts, and other parts controlling tool movement are in working order. However, unless calibrating axis movements with the human-machine interface (HMI), components that control axis movements should generally be serviced by the machine manufacturer.

Some five-axis machines are equipped with a stationary worktable and a nozzle on a movable robotic arm, which can move in all three linear axes and two rotational axes. Five-axis machines allow for more advanced part geometry than three- or four-axis machines. Joints on robotic arms that control precise movements must be maintained to ensure proper function. Damaged or broken components must typically be handled by a robot repair technician.

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## Review: Basic DED Maintenance Considerations

- **Three-axis machine**: Part builds with complex geometries may require cleanup of support materials.
- **Multi-axis machine**: Requires advanced maintenance tasks to ensure the function of linear and rotating components.
- **Powder feed system**: Requires maintenance to ensure the material hopper is free of contaminants.
- **Wire feed system**: May require cleanup of excess lubricant.
- **Predictive maintenance**: Uses data collection and analysis to improve efficiency.

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## Basic Cleaning Methods

Cleaning is part of DED machine maintenance. You must always wear the proper personal protective equipment (PPE) when cleaning the machine, including chemically resistant nitrile gloves.

Typically, you must wipe machine components with a lint-free or microfiber cloth and a mild cleaning solution. This includes wiping the build plate and worktable. On both powder-based and electron beam freeform fabrication (EBFF) machines, be sure to wipe down windows and other surfaces in the build chamber. You must also routinely vacuum the build chamber, material hopper, feed lines, and other areas of the machine. In addition to removing dust and other contaminants, vacuuming material hoppers and feed lines between part builds clears away materials from the previous build that are not compatible with the current build material.

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## Deep Cleaning Methods

Use a wire brush to clean spatter from material feed nozzles, deposition heads, and welding torches. Because metal powders can be combustible, it is best to use an antistatic brush when cleaning powder feed systems to safeguard against the powder potentially igniting.

Sometimes, you may need compressed air to blow free spatter or debris stuck to machine surfaces, or to clear any clogged feed lines. However, pressure levels should be less than 30 pounds per square inch (206,843 Pascals) according to guidelines from the Occupational Safety and Health Administration (OSHA). Always wear proper PPE to protect yourself from respiratory hazards, such as a dust mask or respirator, when cleaning with compressed air.

You may also use ultrasonic cleaning or laser cleaning methods to remove stuck debris and spatter from machine parts. These methods are safer than using compressed air but use more expensive equipment. You must also ensure that the ultrasonic cleaning solution is safe for the component surface. Typically, it is safest to use distilled water and an alkaline cleaning solution, such as a mild detergent or ammonia, to clean metal components with ultrasonic methods.

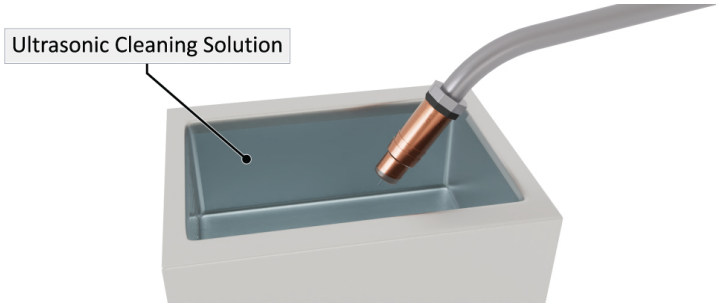


**Figure 1:**

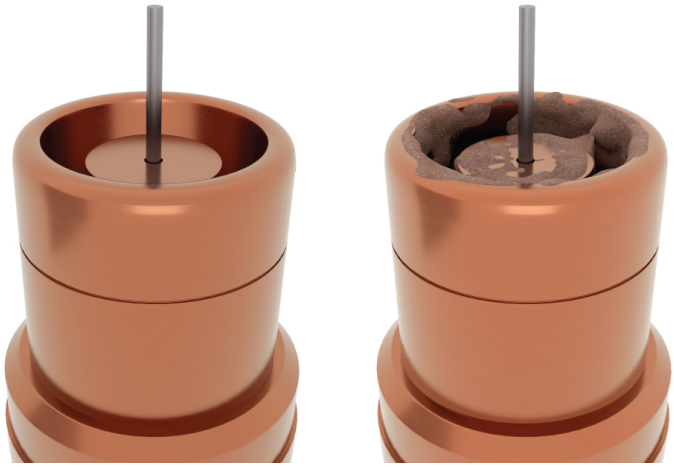

**Figure 2:** Spatter on Weld Nozzle

**Figure 3:** Wire Brush

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## Maintenance Tasks: Checks and Inspections

Machine inspection is a key maintenance task. You must inspect many DED machine components, including powder hoppers or wire feeders, the build chamber, and the worktable, to ensure all mechanical parts are functioning correctly. Visually inspect powder and wire feed lines to ensure there are no obstructions, holes, or tears that could cause material to clog or spill. These issues could also cause shielding gas or carrier gas to leak. Additionally, you should routinely check error codes and other machine notifications before, during, and after operating the machine.

**Figure 1:** Check machine errors and notifications.

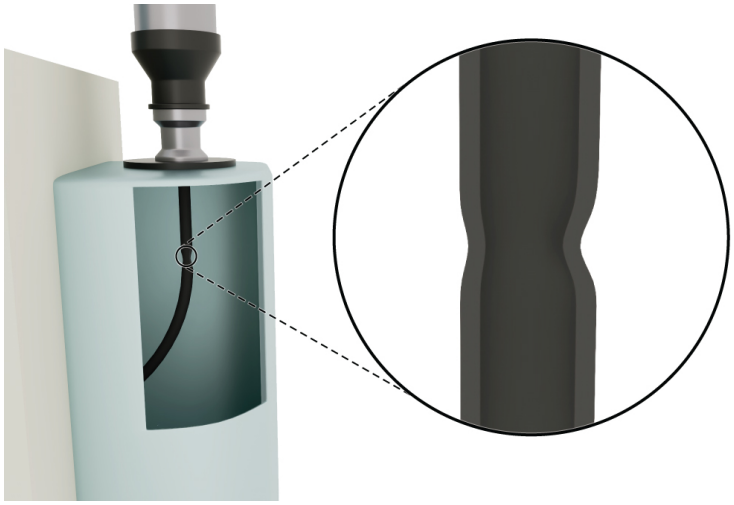

**Figure 2:** Damaged Feed Line

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## Maintenance Tasks: Recording

Documenting, tracking, and analyzing data such as machine hours and repairs can help optimize maintenance and is necessary for PM. PM can greatly reduce machine downtime due to machine failure, malfunctions that lead to recurring print errors, and other unplanned maintenance.

Keeping records of machine errors and error resolution steps over time allows manufacturers to develop best practices and troubleshooting guides. This improves efficiency by providing solutions to common machine issues. Additionally, records of machine component failures, machine repairs, and replaced parts helps you to identify older or worn parts that may require more frequent maintenance and monitoring.

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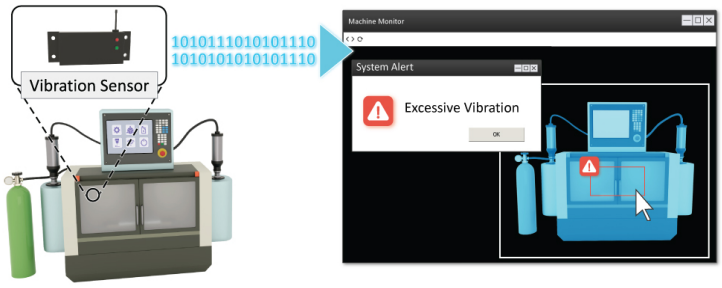
## Maintenance Tasks: Data Capture




PdM requires data capture and data analysis. You may need to install and maintain sensors, such as image sensors and vibration sensors, to capture machine data.

Capturing and analyzing data helps you to determine the root cause of performance issues more effectively and enables PdM strategies. Sensors detect unfavorable or problematic machine conditions, like excessive vibration or heat, and send data to notify you of the issue. You can then analyze the data to decide what maintenance the machine may need. Early warning signs from data analysis can prevent machine failures caused by issues that you may otherwise not detect.

You can also use data for planned maintenance optimization (PMO). PMO customizes PM and PdM tasks to each machine to maximize machine performance and lifespan with the most efficient possible maintenance strategy. PMO efficiency means eliminating unnecessary maintenance steps and costs. For example, a PMO strategy should ensure that preventative maintenance costs do not exceed the cost of replacing the machine.

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## Software System Maintenance

DED maintenance involves not only maintaining the physical condition of a machine, but also maintaining the data and software applications that the machine uses. Since machine connectivity poses a risk of cyber attacks, it is important to establish and maintain cybersecurity.

Establishing passwords and software access levels for each user is key to cybersecurity. Typically, machine operators should have basic access levels that allow them to start and stop printing processes and make basic setup adjustments. However, higher access levels that allow changes to default settings on the machine should be restricted to maintenance technicians.

Software maintenance also involves implementing a plan to protect stored data and maintain data integrity. Unprotected proprietary data can be stolen, so firewalls, local area networks (LANs), and other digital safeguards may be needed to prevent outside access. Faulty data, such as incorrect or incomplete information, can make data analysis inaccurate or unusable, so personnel must routinely check that any data entered is complete with the correct information.

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## Review: Cleaning, Inspection, and Data Maintenance

1. Compressed air is the safest way to remove debris stuck to build chamber surfaces. **FALSE**
2. DED machine inspection may involve checking wire feed lines for obstructions. **TRUE**
3. Machine operators should have access to change default machine settings if needed. **FALSE**
4. Data capture may require maintenance personnel to maintain machine sensors. **TRUE**

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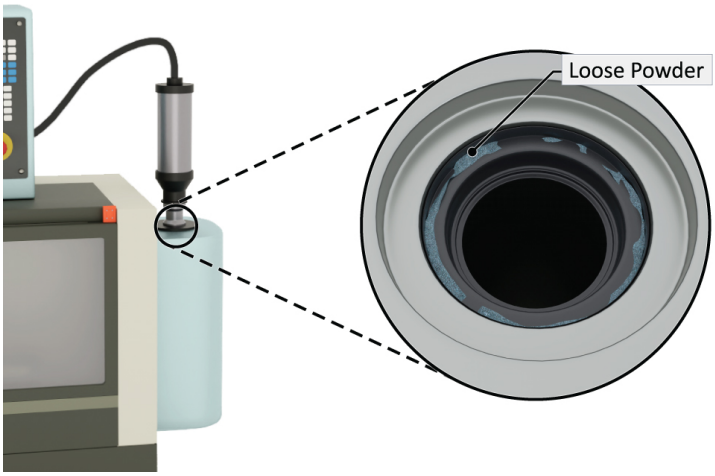
## Regular Maintenance

Your PMO strategy should involve using data analysis to map out daily and weekly DED maintenance tasks.

**Surface grinders can resurface the build plate.**




**Maintenance includes inspecting machine components.**

Daily maintenance tasks for DED might include refilling material hoppers, resurfacing the build plate, and cleaning the nozzle and other machine surfaces before each part build. Safety checks are also an important part of daily maintenance tasks. Use a safety checklist to check for basic safety hazards, such as spills, leaks, or fire hazards.

Weekly maintenance may involve emptying material hoppers in order to inspect inside components and to vacuum and wipe down the inside surfaces. You should also check powder container seals for leaks and ensure that all safety labels are clearly visible.

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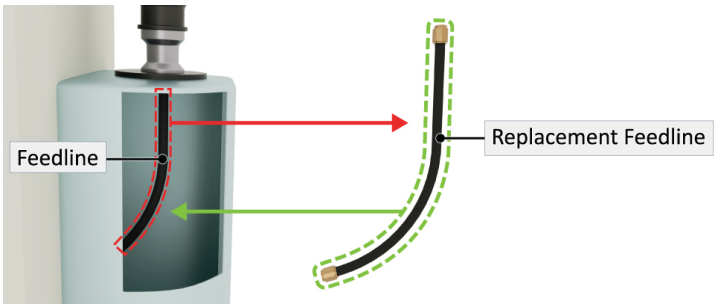
## As-Needed Maintenance




Some maintenance tasks may not be regularly scheduled but instead require attention only under certain circumstances. For instance, you might need to change out feedstock lines between production batches to avoid contamination from different materials.

Additionally, some components may require more attention than others. For example, older components may require weekly inspections, while you may only need to inspect newer components on a monthly basis. Creating a maintenance log can help you keep track of as-needed maintenance.

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## Other Maintenance Considerations

It is important to understand that DED maintenance tasks vary widely based on the machine type and the application. For example, hybrid DED requires additional cleanup to remove chips after machining processes. Chips can cover the workpiece surface, and you must clear them away before adding additional layers to prevent faulty prints.

Always take caution when performing basic maintenance tasks such as cleaning and inspecting parts as you can inadvertently damage or contaminate machine components. Some maintenance tasks should only be done by trained service technicians and not by machine operators. For example, components within a deposition nozzle, such as the lens, require special tools. Always consult the machine operator's manual and follow all guidelines for machine maintenance. Never perform tasks that might void the machine's warranty.

**Figure 1:** Hybrid DED machines require chip removal.

**Figure 2:** Some maintenance is restricted to trained service technicians.

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## Final Review

Which DED maintenance task is likely to be performed on a weekly basis?

- **Inspecting feed components in the material hopper.**
- Cleaning the nozzle or deposition head.
- Resurfacing the build plate.
- Performing safety checks.

How often should DED maintenance technicians perform safety checks? Technicians should perform safety checks:

- **Daily**
- Weekly
- As needed

How often should a technician swap out material feed lines?

- **As needed**
- Daily
- Weekly

How should you determine when to contact the manufacturer for a maintenance task?

- **Refer to the operator's manual.**
- Determine the software access level.
- Consider the age of the machine.

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## Class Vocabulary

**Access Levels** The permissions or restrictions that a user has within a machine or software system. Access levels allow different personnel to use different software functions based on their job role or training.

**Additive Manufacturing** AM. The process of joining or solidifying materials to make an object from a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.

**Build Chamber** The area contained within an additive manufacturing (AM) machine where the part is actually built. Many build chambers are enclosed to improve the safety of the manufacturing operation.

**Build Plate** A removable flat surface on which an additively manufactured part is constructed. Build plates are removable build platforms and must be carefully centered and leveled to ensure a successful build.

**Carrier Gas** An inert gas used to deliver powdered material to the melt pool in a DED system. Carrier gas flow is related to the deposition rate.

**Ceramic** A hard, brittle material that can withstand high temperatures and resist corrosion. Ceramics, which include metal oxides, nitrides, and glasses, are sometimes used as build material in additive manufacturing.

**Chips** A small piece of material that is removed from a workpiece. Chips can get caught between fixture components and the workpiece, reducing finished part accuracy.

**Composites** A group of materials that are made by combining two or more of the following groups: metals, plastics, and ceramics. Composites are sometimes used as build material in some additive manufacturing methods, such as material extrusion and directed energy deposition.

**Compressed Air** Air that has been forced into a small space at high pressure. Compressed air is often used to clean electronics.

**Connectivity** The capability of computer systems and devices to share information across networks. Connectivity helps advance industry but also poses cybersecurity risks.

**Cyber Attacks** An effort to disrupt, disable, or gain illegal access to a digital device or network. Cyber attacks include hacking, phishing, and installing malware.

**Cybersecurity** Protection against criminal or unauthorized access to computer networks, programs, and data. Cybersecurity has become a major industrial concern as networking and connectivity have increased.

**Data Analysis** The process of performing various calculations on organized data sets to gain new information from the data. Data analysis is required for predictive maintenance.

**Data Capture** The process of collecting or reading encoded data and converting it to an electrical signal. Data capture for DED machines is accomplished using sensors.

**Data Integrity** The degree of accuracy and usability of stored data. Data integrity is necessary to produce accurate and usable data insights.

**DED** Directed energy deposition. An additive manufacturing process in which focused thermal energy is used to melt materials as they are deposited. DED is often used with powdered or wire metal feedstock.

**Default Settings** Pre-configured features, options, or functions that an equipment manufacturer sets up prior to a customer receiving the equipment. Default settings on DED machines should only be accessible to trained technicians.

**Deposition Head** A DED machine component that includes the energy source and dispensing nozzle. Deposition heads are used in laser metal deposition (LMD) processes.

**Directed Energy Deposition** DED. An additive manufacturing process that uses focused thermal energy to melt materials as they are deposited. Directed energy deposition is often used with powdered or wire feedstock.

**Dust Mask** A protective nose and mouth cover that filters out some airborne particles. Dust masks are effective at filtering out large particles, such as sand used in binder jetting, but cannot filter out smaller particles, such as nanoscale metal and plastic particles.

**Electron Beam** A narrow stream of electrons that create focused thermal energy. Electron beams are used in some additive manufacturing processes to fuse materials.

**Electron Beam Freeform Fabrication** EBFF. An additive manufacturing technology developed by NASA that uses an electron beam and metal wire feedstock to build and repair parts. EBFF, sometimes referred to as EBF3, uses an

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electron beam and metal wire feedstock .

**Energy Source** The component on a DED machine that melts the material to form the part. DED energy sources include lasers, wire arcs, and electron beams.

**Environmental Controls** Equipment designed to clean or control the environment. Environmental controls can include filtration systems, ventilation systems, or other cleaning devices.

**Error Codes** An alphanumeric message that references a specific cause of malfunction. Error codes appear on the DED machine interface and can be cross-referenced in the manual.

**Feed Lines** Flexible tubes or pipes that deliver feedstock from the feeder to the melt pool in DED systems. Feed lines for DED machines may deliver wire or powder feedstock, depending on the machine type.

**Feed System** The method or mechanism that delivers material to the part build. DED machine feed systems include powder feed and wire feed systems.

**Five-Axis** Machine control that can execute movement along three linear axes and two rotational axes. Five-axis machining capabilities can allow hybrid DED deposition heads to access areas of a part that are difficult to reach otherwise.

**Four-Axis** Machine control that can execute movement along three linear axes and one rotational axis. Four-axis DED machines rotate the workpiece in a fourth axis.

**Gantry** The mechanized bridge on some DED machines that moves the tool in three linear axes. A gantry's movement is controlled by the G code program run by the machine's software.

**Human-Machine Interface** HMI. A device connected to a machine that allows operators to monitor and control a machine or process. A human-machine interface may have a touch screen or control pad.

**Hybrid Manufacturing** A manufacturing approach that produces parts by combining multiple manufacturing processes in a single digital workflow or on a single machine. Hybrid manufacturing may describe a variety of manufacturing process combinations but is most often associated with combining additive and subtractive processes.

**Image Sensors** A device that detects and conveys information used to make an image. Image sensors capture surface data when 3D scanning a part.

**Inert** A gas or other substance that does not chemically react with the substance it contacts. Inert gases include nitrogen, helium, and argon.

**Inspection** Examination of a component or system to ensure that it is operating correctly. Inspection for proactive maintenance involves measuring and comparing various parameters.

**Laser Beam** An intense beam of light that can be precisely aimed and controlled. Lasers are sometimes used to selectively solidify or fuse materials in additive manufacturing methods, including directed energy deposition, powder bed fusion, and vat photopolymerization.

**Laser Cleaning** The use of machines that use laser beams to loosen debris or contaminants on a surface. Laser cleaning is a safer cleaning method than compressed air.

**Linear Axes** An imaginary straight line that describes side-to-side, front-to-back, or up-and-down machine movements. Linear axes typically include the X axis, Y axis, and Z axis.

**Local Area Networks** LANs. A type of wired network that enables communication between a group of computing devices in a specific geographic location. Local area networks are commonly used by manufacturers and other organizations to limit the exchange of digital information to onsite devices and prevent outside access.

**Lubricant** A substance such as oil or grease that is used to prevent friction between two surfaces in relative motion. Lubricant is sometimes used for wire feed systems to reduce friction between the electrode and the feedline.

**Machine Failure** The complete loss of performance in a component or a system. Machine failure can result from electrical and mechanical faults within a system.

**Machining** A manufacturing process that involves removing material to form an object. Machining can occur using traditional methods, like turning, drilling, milling, and grinding, or with less traditional methods that use electricity, heat, or chemical reaction.

**Material Hoppers** A container used to hold raw materials for use in an industrial process. Material hoppers used in metal AM processes hold powdered metal.

**Melt Pool** An area of liquefied material, usually metal. Melt pools are created in directed energy deposition (DED) to fuse powdered metal or metal wire in layers to form a solid part.

**Metal** A naturally occurring material with high strength, hardness, and electrical and thermal conductivity. Metals are categorized as either ferrous or nonferrous.

**Microfiber** A synthetic fiber finer than one denier, commonly used for lint-free cleaning. Microfiber cloths are

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preferred for wiping down components inside a DED machine build chamber.

**Near-Net Shape** A part with dimensions that are close to its final specified form after initial manufacturing steps. Near-net shape parts require little subsequent machining.

**Nitrile** A low-cost copolymer that protects against various chemicals. Nitrile is highly puncture resistant compared to latex.

**Nozzle** The component on a DED machine that dispenses the material. DED nozzles dispense either powder or wire feedstock.

**Occupational Safety And Health Administration** OSHA. A federal agency under the U.S. Department of Labor that ensures safe working conditions and helps to reduce workplace injuries, illnesses, and deaths. The Occupational Safety and Health Administration creates regulations that ensure employees work in a safe and healthy environment.

**Pascals** Pa. A unit of pressure that measures the number of newtons per square meter. A pascal unit is also known as a newton per square meter, which is derived from the International System of Units (SI).

**Passwords** A series of characters, known only by authorized users, that allow the users to access an otherwise locked digital system. Passwords effectively prevent unauthorized access as long as they are not shared or discovered by unauthorized users.

**Personal Protective Equipment** PPE. An article of clothing or a device worn to minimize exposure to hazards and prevent injury. Personal protective equipment may include safety glasses, safety gloves, ear plugs, respirators, or steel-toed boots.

**Planned Maintenance** Maintenance that is performed purposely and regularly in order to prevent machine deterioration or breakdown. Planned maintenance involves performing maintenance on a scheduled basis and helps prevent unscheduled downtime .

**Planned Maintenance Optimization** PMO. A manufacturing improvement method that increases production and reduces waste through continuous attention to the condition of machines and processes. The main goal of planned maintenance optimization is to maximize equipment usefulness across its lifespan.

**PM** Preventative maintenance. Regularly scheduled maintenance tasks performed to prevent unexpected equipment failure. PM can be improved by using data for predictive maintenance.

**PMO** Planned maintenance optimization. A manufacturing improvement method that increases production and reduces waste through continuous attention to the condition of machines and processes. The main goal of PMO is to maximize equipment usefulness across its lifespan.

**Pounds Per Square Inch** psi. A measurement of pressure used in the English system. Pounds per square inch is the pressure resulting from a force of one pound applied to an area of one square inch.

**Powder Feed Systems** A machine or component that delivers powdered material feedstock. Powder feed systems may need to be vacuumed and wiped down during DED maintenance.

**PPE** Personal protective equipment. An article of clothing or a device worn to minimize exposure to hazards and prevent injury. PPE may include safety glasses, safety gloves, ear plugs, respirators, or steel-toed boots.

**Predictive Maintenance** PdM. A maintenance approach that involves collecting and analyzing data in order to service a machine before maintenance issues arise. Predictive maintenance involves performing maintenance when conditions arise that have led to problems in the past, which helps prevent unscheduled downtime.

**Preventative Maintenance** PM. Regularly scheduled maintenance tasks performed to prevent unexpected equipment failure. Preventative maintenance can be improved by using data for predictive maintenance.

**Repeatability** The ability of a machine or process to continually deliver consistent and uniform results. Repeatability is crucial for efficiently producing parts that meet quality requirements.

**Respirator** A breathing device worn to prevent inhalation of hazardous substances. Respirators may purify air or supply air to the wearer.

**Robotic Arm** A programmable or remote-controlled device that simulates the movement of a human arm. Robotic arms are used in a variety of assembly and manufacturing applications, including DED.

**Rotational Axis** An axis that describes turning or rotation around a linear axis. Rotational axes often include the A axis that rotates around the X axis, the B axis that rotates around the Y axis, and the C axis that rotates around the Z axis.

**Seals** An intentional obstruction designed to keep materials, such as powder granules or fluids, inside a system. A seal prevents leakage and contamination.

**Sensors** A device that detects a change in a physical condition and turns it into an electrical signal. Sensors are common analog input devices that respond to variable conditions, such as pressure, temperature, or flow.

**Shielding Gas** A gas that protects the melt pool and energy source from atmospheric contamination. Shielding gas is supplied by a cylinder and flows through welding torch or deposition head on a DED

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machine.

**Spatter** Liquid metal droplets expelled during a welding or DED process. Spatter can leave undesirable dots of metal on a workpiece surface.

**Support Structures** The substance used to create the support structures for an additively manufactured part. Support structure material is often slightly different from the build material so that it can be more easily removed once the part is finished.

**Three-Axis** Machine control that can execute linear movements along two horizontal axes and one vertical axis. Three-axis DED machines can build or repair some basic parts efficiently but are less suitable for more complex parts.

**Troubleshooting** A systematic approach to solving problems quickly and efficiently. Troubleshooting often involves a logical process of elimination to identify the source of a problem.

**Ultrasonic Cleaning** The use of machines that use ultrasonic sound waves to loosen debris or contaminants on a surface. Ultrasonic cleaning is a safer cleaning method than compressed air.

**Vacuum-Sealed Chamber** A sealed enclosure from which air is removed by a vacuum pump. Vacuum-sealed chambers surround the entire build area in some DED systems.

**Vibration Sensors** A smart device that monitors machine component vibration levels. Vibration sensors can detect hazardous machine conditions.

**Warranty** A guarantee provided by the manufacturer covering repair or replacement of defective machine components within a specified time.

**Welding Torch** The instrument used to generate the arc or flame for welding. The welding torch may hold an electrode or release gases that produce a flame.

**Wire Arc** The area in which electricity jumps from an electrode to a conductor to produce extreme heat and light. Wire arcs are used in thermal welding and cutting applications, as well as in directed energy deposition.

**Wire Feed Systems** A machine or component that delivers material feedstock in the form of a wire spool. Wire feed systems may need to be cleaned during DED maintenance.

**Wire Feeder** A device that delivers a wire electrode to the melt pool. The wire feeder may be built inside the power source or be an external device set beside it.

**Workpiece** A part that is in the process of being manufactured. A workpiece may be a complete product by itself or one component of a product consisting of many parts.

**Worktable** The part of a DED machine that supports the workpiece. The worktable, or table, can move along a combination of linear and rotational axes, depending on the configuration of the machine.

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